

B2 Symmetry & Relativity — Model Solutions, 2018

Question 1 — 4-vectors, kinematics, dynamics

(a) Intervals, null separation, orthogonality

Problem. Conditions for spacelike/timelike intervals between events $D = (ct_d, \mathbf{x}_d)$, $G = (ct_g, \mathbf{x}_g)$. Null separation? Conditions $Y^\mu X_\mu = 0$. Which pairs of A, K, J are orthogonal?

Interval 4-vector:

$$\Delta X^\mu = D^\mu - G^\mu = (c\Delta t, \Delta \mathbf{x}), \quad \Delta t = t_d - t_g, \quad \Delta \mathbf{x} = \mathbf{x}_d - \mathbf{x}_g.$$

Minkowski-square (signature $(+, -, -, -)$):

$$\Delta X^\mu \Delta X_\mu = c^2 \Delta t^2 - |\Delta \mathbf{x}|^2.$$

Timelike (causally connected by sub-light signal):

$$c^2 \Delta t^2 - |\Delta \mathbf{x}|^2 > 0$$

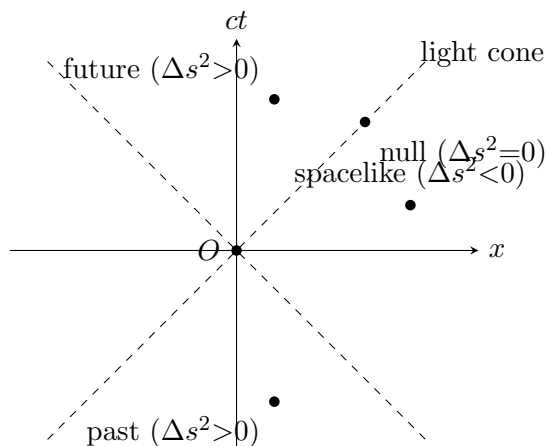
Spacelike (no causal signal):

$$c^2 \Delta t^2 - |\Delta \mathbf{x}|^2 < 0$$

Null (light-like, connected by photon):

$$c^2 \Delta t^2 - |\Delta \mathbf{x}|^2 = 0$$

Yes, two events can be connected by a null vector — a photon emitted at G and absorbed at D traces such a null worldline.



When is $Y^\mu X_\mu = 0$? Three independent cases:

1. One of the 4-vectors vanishes: $Y^\mu = 0$ or $X^\mu = 0$.
2. Both are null and parallel: $Y^\mu Y_\mu = X^\mu X_\mu = 0$ with $Y^\mu \propto X^\mu$.
3. One is timelike (or null) and the other spacelike (or null), oriented so the spatial overlap cancels the temporal product — i.e. a genuinely orthogonal Minkowski pair.

A timelike 4-vector cannot be orthogonal to another timelike 4-vector (their inner product retains the sign of $c^2 \Delta t^2$).

Pairs among A, K, J .

4-acceleration vs. 4-velocity satisfies $A^\mu U_\mu = 0$ identically (differentiate $U^\mu U_\mu = c^2$).

For a charged particle of rest charge density ρ_0 , $J^\mu = \rho_0 U^\mu \propto U^\mu$. Hence

$$A^\mu J_\mu = \rho_0 A^\mu U_\mu = 0.$$

$A \cdot J = 0 \text{ always.}$

For K (photon, null) versus A (spacelike): generically $A^\mu K_\mu \neq 0$ unless special alignment. Likewise $K \cdot J \neq 0$ in general (a charge moving relative to the photon will see a nonzero phase rate). So only $A \cdot J = 0$ is guaranteed.

(b) 4-momentum, 4-acceleration, parallel case

Problem. Define $\mathbf{P}, \mathbf{A}$. Compute $P^\mu P_\mu$. Show $A^\mu A_\mu = \gamma^6 a^2$ when $\mathbf{v} \parallel \mathbf{a}$.

Definitions:

$$P^\mu = mU^\mu = (\gamma mc, \gamma m\mathbf{v}), \quad A^\mu = \frac{dU^\mu}{d\tau} = \gamma \frac{dU^\mu}{dt}.$$

Compute $P^\mu P_\mu$:

$$P^\mu P_\mu = \gamma^2 m^2 c^2 - \gamma^2 m^2 v^2.$$

Factor $\gamma^2 m^2$:

$$P^\mu P_\mu = \gamma^2 m^2 (c^2 - v^2).$$

Use $\gamma^2 (c^2 - v^2) = c^2$:

$P^\mu P_\mu = m^2 c^2.$

4-acceleration with $\mathbf{v} \parallel \mathbf{a}$. Spatial 4-velocity $\mathbf{u} = \gamma \mathbf{v}$. Then

$$\frac{du^0}{dt} = c \dot{\gamma}, \quad \frac{d\mathbf{u}}{dt} = \dot{\gamma} \mathbf{v} + \gamma \mathbf{a}.$$

Compute $\dot{\gamma}$ for $\mathbf{v} \parallel \mathbf{a}$:

$$\dot{\gamma} = \frac{d}{dt}[(1 - v^2/c^2)^{-1/2}] = \gamma^3 \frac{\mathbf{v} \cdot \mathbf{a}}{c^2} = \gamma^3 \frac{va}{c^2}.$$

Then

$$A^0 = \gamma \dot{\gamma} c = \gamma^4 \frac{va}{c}, \quad \mathbf{A} = \gamma(\dot{\gamma} \mathbf{v} + \gamma \mathbf{a}) = \gamma^4 \frac{v^2 a}{c^2} \hat{\mathbf{a}} + \gamma^2 a \hat{\mathbf{a}}.$$

Simplify $|\mathbf{A}|$:

$$|\mathbf{A}| = \gamma^2 a \left(\gamma^2 \frac{v^2}{c^2} + 1 \right) = \gamma^2 a \cdot \gamma^2 = \gamma^4 a.$$

Used $1 + \gamma^2 v^2/c^2 = \gamma^2$. Compute the invariant:

$$A^\mu A_\mu = (A^0)^2 - |\mathbf{A}|^2 = \gamma^8 \frac{v^2 a^2}{c^2} - \gamma^8 a^2.$$

Factor:

$$A^\mu A_\mu = \gamma^8 a^2 \left(\frac{v^2}{c^2} - 1 \right) = -\gamma^8 a^2 / \gamma^2.$$

$$\boxed{A^\mu A_\mu = -\gamma^6 a^2.}$$

The magnitude $|A^\mu A_\mu| = \gamma^6 a^2$ as required (sign reflects A being spacelike).

(c) Inverse-square central force: energy conservation

Problem. For $\mathbf{f} = \alpha \mathbf{r} / r^3$, derive $\gamma m c^2 - \alpha / r = \text{const.}$

Relativistic 2nd law (rest mass preserved):

$$\frac{d}{dt}(\gamma m \mathbf{v}) = \mathbf{f}.$$

Energy equation:

$$\frac{d}{dt}(\gamma m c^2) = \mathbf{f} \cdot \mathbf{v}.$$

Insert force:

$$\mathbf{f} \cdot \mathbf{v} = \frac{\alpha}{r^3} \mathbf{r} \cdot \mathbf{v}.$$

Identify $\mathbf{r} \cdot \mathbf{v} = r \dot{r}$:

$$\mathbf{f} \cdot \mathbf{v} = \frac{\alpha \dot{r}}{r^2}.$$

Rewrite as total derivative:

$$\frac{\alpha \dot{r}}{r^2} = -\frac{d}{dt} \left(\frac{\alpha}{r} \right).$$

Combine:

$$\frac{d}{dt} \left(\gamma m c^2 + \frac{\alpha}{r} \right) = 0.$$

Hence (for attractive force $\alpha < 0$, or the sign convention with potential $-\alpha/r$):

$$\boxed{\gamma m c^2 - \frac{\alpha}{r} = \text{const.}}$$

Example. Coulomb attraction between an electron and a proton: $\alpha = -e^2/(4\pi\epsilon_0)$ (or gravitational $\alpha = -GMm$ in the weak-field Newtonian limit).

(d) Electron accelerated across a 5 m gap, $E = 10 \text{ MV/m}$

Problem. Find γ at exit, time of flight, sketch $\gamma(t), \beta(t), \Delta\beta$ if field reduced 20%, and proton case.

Energy gain (work-energy theorem):

$$\gamma m c^2 = m c^2 + e E L.$$

Solve for γ :

$$\gamma = 1 + \frac{e E L}{m c^2}.$$

Insert numbers ($m c^2 = 0.511 \text{ MeV}$, $e E L = 10 \text{ MeV/m} \times 5 \text{ m} = 50 \text{ MeV}$):

$$\gamma = 1 + \frac{50}{0.511} = 1 + 97.85.$$

$$\boxed{\gamma \approx 98.85}$$

Time of flight. Equation of motion (constant force, 1-D):

$$\frac{d}{dt}(\gamma mv) = eE.$$

Integrate from rest:

$$\gamma mv = eEt.$$

Square and use $\gamma^2 v^2 = c^2(\gamma^2 - 1)$:

$$m^2 c^2 (\gamma^2 - 1) = (eEt)^2.$$

Solve for $\gamma(t)$:

$$\gamma(t) = \sqrt{1 + \left(\frac{eEt}{mc}\right)^2}.$$

Position from $dx/dt = v = c\sqrt{1 - 1/\gamma^2}$:

$$x(t) = \frac{mc^2}{eE} \left[\sqrt{1 + (eEt/mc)^2} - 1 \right].$$

Set $x = L$ and solve:

$$\sqrt{1 + (eEt/mc)^2} = 1 + \frac{eEL}{mc^2} = \gamma.$$

Square:

$$\left(\frac{eEt}{mc}\right)^2 = \gamma^2 - 1.$$

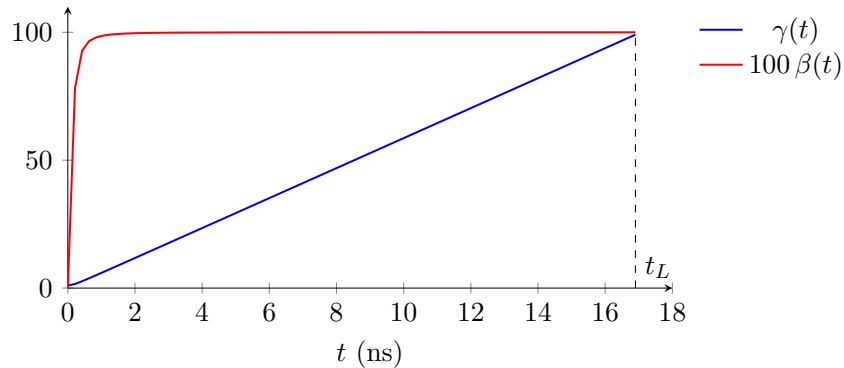
Solve t :

$$t = \frac{mc}{eE} \sqrt{\gamma^2 - 1}.$$

Numerical: $\sqrt{\gamma^2 - 1} = \sqrt{98.85^2 - 1} \approx 98.84$, $mc^2/(eE) = 0.511 \text{ MeV}/10 \text{ MV/m} = 0.0511 \text{ m}$, so $mc/(eE) = 0.0511 \text{ m}/c$:

$$t = \frac{0.0511 \text{ m}}{c} \times 98.84 = \frac{5.05 \text{ m}}{c}.$$

$$\boxed{t \approx 1.69 \times 10^{-8} \text{ s} \approx 16.9 \text{ ns.}}$$



Field reduced by 20% ($E \rightarrow 0.8E$). New $\gamma' = 1 + 0.8 \times 97.85 = 79.28$. Both β and β' are very close to 1:

$$1 - \beta = \frac{1}{2\gamma^2} = \frac{1}{2 \times 98.85^2} = 5.12 \times 10^{-5}, \quad 1 - \beta' = \frac{1}{2 \times 79.28^2} = 7.96 \times 10^{-5}.$$

$$\Delta\beta \approx -2.8 \times 10^{-5} \text{ (decrease).}$$

Proton instead. $m_p c^2 = 938.3 \text{ MeV} \gg 50 \text{ MeV}$, so non-relativistic:

$$\gamma_p = 1 + \frac{50}{938.3} = 1.053.$$

$$\beta_p = \sqrt{1 - 1/\gamma_p^2} = 0.314.$$

$$\gamma_p \approx 1.05, \beta_p \approx 0.31.$$

Question 2 — Photons, decays, fields, mirror reflection

(a) Photon 4-wavevector, phase invariance

Problem. Components of $\mathbf{K}$, dispersion, v_{ph} and v_g , show ϕ is Lorentz-invariant; is v_{ph} invariant? 4-wavevector:

$$K^\mu = (\omega/c, \mathbf{k}), \quad \omega = c|\mathbf{k}|.$$

Null condition (vacuum):

$$K^\mu K_\mu = \omega^2/c^2 - |\mathbf{k}|^2 = 0.$$

Phase velocity (along $\hat{\mathbf{k}}$):

$$v_{\text{ph}} = \omega/|\mathbf{k}| = c.$$

Group velocity:

$$\mathbf{v}_g = \nabla_{\mathbf{k}} \omega = c\hat{\mathbf{k}}, \quad |\mathbf{v}_g| = c.$$

Phase of plane wave:

$$\phi(\mathbf{r}, t) = \mathbf{k} \cdot \mathbf{r} - \omega t = -K^\mu X_\mu.$$

A contraction of two 4-vectors is a Lorentz scalar:

$$\phi = -K^\mu X_\mu \text{ is Lorentz-invariant.}$$

Is v_{ph} Lorentz-invariant? For a vacuum photon, $\omega'/|\mathbf{k}'| = c$ in every frame, so yes — but only because c is the universal speed limit. In a medium with $v_{\text{ph}} \neq c$, a Lorentz boost generally changes both ω and $|\mathbf{k}|$ such that v_{ph} is *not* invariant. So:

$$v_{\text{ph}} = c \text{ in vacuum: invariant. In a medium: not invariant.}$$

(b) Forbidden and allowed photon processes

Problem. Can $\gamma \rightarrow m + \gamma'$? Photon-photon $\rightarrow e^+e^-$: minimum number, threshold energy.

Decay $\gamma \rightarrow m + \gamma'$. 4-momentum conservation:

$$K_1^\mu = P_m^\mu + K_2^\mu.$$

Square (use $K_1^\mu K_{1\mu} = K_2^\mu K_{2\mu} = 0$, $P_m^\mu P_{m\mu} = m^2 c^2$):

$$0 = m^2 c^2 + 2P_m^\mu K_{2\mu}.$$

Solve:

$$P_m^\mu K_{2\mu} = -\frac{1}{2}m^2 c^2.$$

In rest frame of m : $P_m = (mc, \mathbf{0})$, $K_2 = (\omega_2/c, \mathbf{k}_2)$, so $P \cdot K_2 = m\omega_2 \geq 0$, contradicting the negative requirement (with metric $(+, -, -, -)$ and a real positive ω_2). Equivalently, work in the m rest frame: 3-momentum requires $\mathbf{k}_1 = \mathbf{k}_2$, but then $\omega_1 = \omega_2$ and energy conservation gives $mc^2 = 0$.

Forbidden: a single photon cannot decay into $m + \gamma$.

Photon-photon $\rightarrow e^+e^-$. A single photon has $K^\mu K_\mu = 0$ and cannot create a massive system ($P^2 = (2m_e c)^2 > 0$) by itself. Two photons suffice if they are not collinear:

$$(K_1 + K_2)^\mu (K_1 + K_2)_\mu = 2K_1^\mu K_{2\mu} \geq (2m_e c)^2.$$

Minimum: 2 photons.

Threshold. Equal frequencies, head-on $\mathbf{k}_1 = -\mathbf{k}_2$, $|\mathbf{k}_1| = |\mathbf{k}_2| = \omega/c$:

$$K_1^\mu K_{2\mu} = \omega^2/c^2 - \mathbf{k}_1 \cdot \mathbf{k}_2 = \omega^2/c^2 + \omega^2/c^2 = 2\omega^2/c^2.$$

Threshold (e^+e^- produced at rest in CM):

$$2 \cdot 2\omega^2/c^2 = (2m_e c)^2.$$

Solve:

$$\omega = m_e c^2 / \hbar.$$

Photon energy:

$$E_\gamma = \hbar\omega = m_e c^2 = 0.511 \text{ MeV}.$$

$$E_\gamma^{\min} = m_e c^2 \approx 0.511 \text{ MeV per photon}.$$

(c) Conserved 4-momentum components in $\mathbf{E}$ or $\mathbf{B}$ field

Problem. Electron with $\mathbf{v}_0 = (v_0, 0, 0)$. Which P^μ are conserved for (i) $\mathbf{E} = (E_x, 0, 0)$, (ii) $\mathbf{B} = (0, B_y, 0)$.

Lorentz force:

$$\frac{d\mathbf{p}}{dt} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \frac{dE}{dt} = e\mathbf{E} \cdot \mathbf{v}.$$

(i) $\mathbf{E} = (E_x, 0, 0)$, **no** $\mathbf{B}$. The force is $eE_x \hat{x}$:

$$\frac{dp_y}{dt} = 0, \quad \frac{dp_z}{dt} = 0.$$

Energy is *not* conserved ($\mathbf{E} \cdot \mathbf{v} = E_x v_x \neq 0$). So

$$P^y, P^z \text{ conserved; } P^0, P^x \text{ not.}$$

Initial conditions give $p_y = p_z = 0$ throughout.

(ii) $\mathbf{B} = (0, B_y, 0)$, **no** $\mathbf{E}$. Magnetic force does no work:

$$\frac{dE}{dt} = 0 \Rightarrow P^0 \text{ conserved.}$$

$\mathbf{v} \times \mathbf{B}$ has no y -component when $\mathbf{v}$ has no y -component (initially $\mathbf{v} = (v_0, 0, 0)$). Check that v_y remains zero: starting $\mathbf{v} = (v_x, 0, v_z)$ stays in the xz -plane:

$$\mathbf{v} \times \mathbf{B} = (v_x, 0, v_z) \times (0, B_y, 0) = (-v_z B_y, 0, v_x B_y).$$

No y -component generated, so p_y stays zero — conserved.

$$P^0 \text{ and } P^y \text{ conserved; } P^x, P^z \text{ rotate (circular motion).}$$

(d) Reflection from a relativistic mirror

Problem. Mirror moves at $\beta_x = 0.99$ along $\hat{x}$. Lab photon $\lambda_1 = 1 \mu\text{m}$. In mirror frame $\mathbf{k}' = (-k'_x, k'_y, 0)$ with $|k'_x| = |k'_y|$. Find lab reflection angle and reflected λ_2 .

Step 1: Lorentz factor. $\gamma = 1/\sqrt{1 - 0.99^2} = 7.0888$.

Step 2: Lab incident $\mathbf{K}_1$. In mirror frame, $|k'_x| = |k'_y|$ with $k'_x < 0$; the photon moves in $-\hat{x}$ direction (against mirror velocity), as stated:

$$K_1'^\mu = (\omega'/c) (1, -1/\sqrt{2}, 1/\sqrt{2}, 0), \quad \omega' = c\sqrt{k_x'^2 + k_y'^2} = c|k'_x|\sqrt{2}.$$

Boost to lab (frame S moves with mirror velocity $+v\hat{x}$ relative to S' , so S' at velocity $+v$ relative to S ; the inverse boost from $S' \rightarrow S$ uses $-v$, i.e. $K^\mu = \Lambda^\mu_\nu(+v)K'^\nu$):

$$\omega_1/c = \gamma(\omega'/c + \beta k'_x) = \gamma(\omega'/c)(1 - \beta/\sqrt{2}).$$

$$k_{1x} = \gamma(k'_x + \beta\omega'/c) = \gamma(\omega'/c)(\beta - 1/\sqrt{2}).$$

$$k_{1y} = k'_y = (\omega'/c)/\sqrt{2}.$$

Numerical with $\beta = 0.99$, $\gamma = 7.0888$, $1/\sqrt{2} = 0.7071$:

$$\omega_1/c = 7.0888 (\omega'/c)(1 - 0.7000) = 2.127 \omega'/c.$$

But we want to use the given $\lambda_1 = 1 \mu\text{m}$, so this fixes ω' :

$$\omega' = \omega_1/2.127.$$

Step 3: Reflection in mirror frame. A perfect mirror flips the normal component of $\mathbf{k}'$:

$$K_2'^\mu = (\omega'/c) (1, +1/\sqrt{2}, 1/\sqrt{2}, 0).$$

Reflection angle in mirror frame is 45° from normal.

Step 4: Boost reflected photon back to lab.

$$\omega_2/c = \gamma(\omega'/c + \beta k'_{2x}) = \gamma(\omega'/c)(1 + \beta/\sqrt{2}).$$

$$k_{2x} = \gamma(k'_{2x} + \beta\omega'/c) = \gamma(\omega'/c)(1/\sqrt{2} + \beta).$$

$$k_{2y} = k'_{2y} = (\omega'/c)/\sqrt{2}.$$

Numerics:

$$\omega_2/c = 7.0888 (\omega'/c)(1 + 0.7071) = 12.107 \omega'/c.$$

Frequency ratio:

$$\omega_2/\omega_1 = 12.107/2.127 = 5.692.$$

Wavelength of reflected photon:

$$\lambda_2 = \lambda_1 \omega_1/\omega_2 = 1 \mu\text{m}/5.692.$$

$\lambda_2 \approx 0.176 \mu\text{m} \approx 176 \text{ nm}.$

Reflection angle in lab. Measure from $+\hat{x}$ (mirror normal direction):

$$\tan \theta_2 = k_{2y}/k_{2x} = \frac{1/\sqrt{2}}{\gamma(1/\sqrt{2} + \beta)} = \frac{0.7071}{7.0888 \times 1.6971}.$$

Compute denominator:

$$7.0888 \times 1.6971 = 12.030.$$

So

$$\tan \theta_2 = 0.7071/12.030 = 0.05878.$$

$$\theta_2 = \arctan(0.0588) \approx 3.36^\circ$$

The reflected photon emerges almost along $+\hat{x}$ (relativistic beaming towards mirror motion direction).

Question 3 — Bounds, Compton, periodic-field undulator

(a) Bounds on forces, v_{ph} ; Compton shift

Problem. Bounds on forces/accelerations? On phase velocity? Derive Compton shift.

No upper bound on force or acceleration. Special relativity bounds the *velocity* of a massive particle ($v < c$), but $\mathbf{a} = d\mathbf{v}/dt$ can be arbitrarily large in transient regimes; only the integrated speed-gain is constrained. For force, $\mathbf{f} = d\mathbf{p}/dt$ is similarly unbounded.

Phase velocity is unbounded. $v_{ph} = \omega/|\mathbf{k}|$ in a medium can exceed c (e.g. X-rays in matter, plasma above ω_p). The bound is on the group velocity / signal propagation, not phase.

Compton shift. Initial: photon $K = (\omega/c)(1, \hat{n})$, electron at rest $P_e = (m_e c, \mathbf{0})$. After: photon $K' = (\omega'/c)(1, \hat{n}')$, electron P'_e . Conservation:

$$K + P_e = K' + P'_e.$$

Rearrange and square:

$$(P_e + K - K')^\mu (P_e + K - K')_\mu = P'^\mu_e P'_{e\mu}.$$

Use $P_e^2 = P'^2_e = m_e^2 c^2$, $K^2 = K'^2 = 0$:

$$m_e^2 c^2 + 2P_e \cdot K - 2P_e \cdot K' - 2K \cdot K' = m_e^2 c^2.$$

Cancel $m_e^2 c^2$:

$$P_e \cdot (K - K') = K \cdot K'.$$

Evaluate (rest frame $P_e = (m_e c, \mathbf{0})$):

$$P_e \cdot K = m_e \omega, \quad P_e \cdot K' = m_e \omega', \quad K \cdot K' = (\omega \omega' / c^2)(1 - \cos \theta).$$

Substitute:

$$m_e(\omega - \omega') = (\omega \omega' / c^2)(1 - \cos \theta).$$

Divide by $m_e \omega \omega'$:

$$\frac{1}{\omega'} - \frac{1}{\omega} = \frac{1 - \cos \theta}{m_e c^2}.$$

Convert to wavelength $\lambda = 2\pi c / \omega$:

$$\lambda' - \lambda = \frac{2\pi \hbar}{m_e c} (1 - \cos \theta) = \lambda_C (1 - \cos \theta)$$

with Compton wavelength $\lambda_C = \hbar / m_e c \approx 2.43 \text{ pm}$.

(b) Distance between two photons in S'

Problem. Two photons separated by x_0 along $\hat{x}$ in S . Find separation in S' moving with $\mathbf{v} = (v_x, 0, 0)$.

The two photons are at events $E_1 = (ct_0, x_0, 0, 0)$ and $E_2 = (ct_0, 0, 0, 0)$ at lab time t_0 . In S' these events do not occur simultaneously; the photons are points propagating along null worldlines, so we need the spatial separation between them in S' measured at a single S' -time.

A photon has worldline $x = x_a + ct$ (forward-moving); both photons have parallel null worldlines $x_1 = x_0 + ct$, $x_2 = ct$. The separation $x_1 - x_2 = x_0$ is constant in S . Apply Lorentz transformation: at fixed t' , find $\Delta x'$.

For each photon $K^\mu = (\omega_a/c)(1, 1, 0, 0)$, the photon worldline parametrically transforms; in S' (boost $+v$):

$$x'_a = \gamma(x_a - vt), \quad t'_a = \gamma(t - vx_a/c^2).$$

Solve t_a at fixed t' : $t = (t' + vx_a/c^2)/\gamma$. Substitute:

$$x'_a = \gamma(x_a - v(t' + vx_a/c^2)/\gamma) = \gamma x_a(1 - v^2/c^2) - vt' = x_a/\gamma - vt'.$$

Difference:

$$\Delta x' = (x_{1,a} - x_{2,a})/\gamma \Big|_{\text{at fixed } t'}.$$

But this measures separation along the photons' shared null trajectory at constant lab time. The cleaner way: the photons' worldline parameter x_a (lab-frame x -intercept) is Lorentz-mapped via the Doppler factor for null separations.

Use the invariant interval. $\Delta X^\mu = (0, x_0, 0, 0)$ between simultaneous S -events:

$$\Delta X'^0 = -\gamma\beta x_0, \quad \Delta X'^1 = \gamma x_0.$$

The events are not simultaneous in S' ($\Delta t' = -\gamma\beta x_0/c \neq 0$). To get the S' -spatial separation between the photons at a single S' -time, slide one photon along its null worldline by $\Delta t'$:

$$\Delta x'_{\text{measured}} = \gamma x_0 - c \Delta t' = \gamma x_0 - c(-\gamma\beta x_0/c) = \gamma x_0(1 + \beta).$$

$$\Delta x' = x_0 \gamma(1 + \beta) = x_0 \sqrt{\frac{1 + \beta}{1 - \beta}}.$$

This is the relativistic Doppler factor for a separation along the propagation direction — consistent with photons being "stretched apart" when the observer recedes.

(c) Head-on Compton: $\lambda_2 \approx \lambda_1/(4\gamma^2)$

Problem. Electron and photon counter-propagate; head-on backscatter. Show $\lambda_2 \approx \lambda_1/(4\gamma^2)$ when $\hbar\omega \ll m_e c^2$. Numerical for $\beta = 0.999$, $\lambda_1 = 8 \mu\text{m}$.

Method: Doppler shift twice. In electron rest frame the Compton shift on a low-energy photon is negligible (because $\hbar\omega' \ll m_e c^2$): $\omega'_{\text{out}} \approx \omega'_{\text{in}}$.

Doppler shift lab $\rightarrow$ electron rest frame (head-on collision, photon moves $-\hat{x}$, electron at $+\hat{x}$ velocity):

$$\omega' = \gamma(1 + \beta)\omega_1.$$

In rest frame, backscatter: outgoing photon at $+\hat{x}$. Doppler back to lab:

$$\omega_2 = \gamma(1 + \beta)\omega'.$$

Combine:

$$\omega_2 = \gamma^2(1 + \beta)^2 \omega_1.$$

Ultra-relativistic: $1 + \beta \rightarrow 2$, $1 - \beta \approx 1/(2\gamma^2)$:

$$\omega_2 \approx 4\gamma^2 \omega_1.$$

Wavelength:

$$\lambda_2 \approx \lambda_1/(4\gamma^2).$$

Numerical. $\beta = 0.999$:

$$\gamma = 1/\sqrt{1 - 0.999^2} = 1/\sqrt{0.001999} = 22.366.$$

$$4\gamma^2 = 4 \times 500.2 = 2000.8.$$

$$\lambda_2 = 8 \mu\text{m}/2000.8.$$

$$\lambda_2 \approx 4.0 \text{ nm}.$$

(d) Plane EM wave: A^μ , $\mathbf{E}$, $\mathbf{B}$

Problem. Plane wave ω , $\mathbf{k} = (k_x, 0, 0)$. Write a possible A^μ , find $\mathbf{E}$, $\mathbf{B}$.

Lorenz-gauge null 4-potential, transverse polarization along $\hat{\mathbf{y}}$:

$$A^\mu = (0, 0, A_0 \cos(k_x x - \omega t), 0).$$

Fields:

$$\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A} = -\partial_t A_0 \hat{\mathbf{y}} = -A_0 \omega \sin(k_x x - \omega t) \hat{\mathbf{y}}.$$

Use $\omega = ck_x$:

$$\mathbf{E} = -A_0 ck_x \sin(k_x x - \omega t) \hat{\mathbf{y}}.$$

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Only A_y depends on x :

$$\mathbf{B} = (\partial_x A_y) \hat{\mathbf{z}} = -A_0 k_x \sin(k_x x - \omega t) \hat{\mathbf{z}}.$$

$$\mathbf{E} = -A_0 ck_x \sin(k_x x - \omega t) \hat{\mathbf{y}}, \quad \mathbf{B} = -A_0 k_x \sin(k_x x - \omega t) \hat{\mathbf{z}}$$

$|\mathbf{E}| = c|\mathbf{B}|$, and $\mathbf{E} \times \mathbf{B}$ along $+\hat{\mathbf{x}}$ as required.

(e) Periodic-field undulator: lab vs. electron rest frame

Problem. Undulator $A^\mu = (\phi/c, \mathbf{0})$, $\phi = \phi_0 \cos(k_u z)$. Electron with $\mathbf{v} = (0, 0, v_z)$, ultra-relativistic. Find $\mathbf{E}$, $\mathbf{B}$ in lab and rest frame; is the field in the rest frame an EM wave?

Lab frame.

$$\mathbf{E} = -\nabla\phi = \phi_0 k_u \sin(k_u z) \hat{\mathbf{z}}, \quad \mathbf{B} = \nabla \times \mathbf{A} = \mathbf{0}.$$

$$\mathbf{E}_{\text{lab}} = \phi_0 k_u \sin(k_u z) \hat{\mathbf{z}}, \quad \mathbf{B}_{\text{lab}} = \mathbf{0}.$$

A purely electrostatic, longitudinal, z -periodic field.

Electron rest frame ($z' = \gamma(z - vt)$, $t' = \gamma(t - vz/c^2)$). Boost the field strength tensor. Boost direction parallel to $\mathbf{E}$, so the longitudinal E_z is unchanged and a transverse $\mathbf{B}$ does not appear:

$$E'_z = E_z, \quad B'_\perp = -\gamma\beta (\mathbf{E} \times \hat{\mathbf{v}})_\perp / c = 0.$$

So $\mathbf{B}' = 0$ (boost along $\mathbf{E}$), but the spatial argument transforms:

$$k_u z = k_u (\gamma z' + \gamma v t') = \gamma k_u z' + \gamma k_u v t'.$$

Define $k'_u = \gamma k_u$, $\omega' = \gamma k_u v \approx \gamma k_u c$ (ultra-relativistic):

$$\mathbf{E}'(\mathbf{r}', t') = \phi_0 k_u \sin(k'_u z' + \omega' t') \hat{\mathbf{z}}'.$$

$$\boxed{\mathbf{E}' = \phi_0 k_u \sin(\gamma k_u z' + \gamma k_u v t') \hat{\mathbf{z}}', \quad \mathbf{B}' = \mathbf{0}.}$$

Is this an EM wave? A vacuum plane EM wave satisfies $|\mathbf{E}| = c|\mathbf{B}|$ with $\mathbf{E} \perp \mathbf{k}$, $\mathbf{B} \perp \mathbf{k}$, $\mathbf{B} \neq 0$. Here $\mathbf{B}' = 0$ and $\mathbf{E}' \parallel \hat{\mathbf{z}}'$ (longitudinal to the propagation direction $\hat{\mathbf{z}}'$).

$$\boxed{\text{Not an EM wave: longitudinal } \mathbf{E}, \mathbf{B}' = \mathbf{0}.}$$

It is a longitudinal oscillating electric field — the rest-frame view of a static periodic potential as seen by a co-moving observer (no radiation; it is not transverse).

Question 4 — Polar/axial vectors, $F^{\mu\nu}$, capacitor

(a) Polar vs. axial; $\mathbf{E}, \mathbf{B}$

Problem. Define polar/axial 3-vectors. Are $\mathbf{E}, \mathbf{B}$ polar or axial? Justify via Lorentz force.

Polar vector: flips sign under spatial inversion $\mathbf{r} \rightarrow -\mathbf{r}$ (e.g. $\mathbf{r}, \mathbf{v}, \mathbf{p}, \mathbf{E}$).

Axial (pseudo-) vector: unchanged under inversion (e.g. $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, $\mathbf{B}$, angular velocity).

Lorentz force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Under parity:

$$\mathbf{F} \rightarrow -\mathbf{F}, \quad q\mathbf{E} \rightarrow -q\mathbf{E} \Rightarrow \mathbf{E} \text{ is polar.}$$

$$\mathbf{v} \times \mathbf{B} \rightarrow (-\mathbf{v}) \times \mathbf{B} \rightarrow -\mathbf{v} \times \mathbf{B} \quad \text{requires} \quad \mathbf{B} \rightarrow +\mathbf{B}.$$

So

$$\boxed{\mathbf{E} \text{ polar, } \mathbf{B} \text{ axial.}}$$

(b) 4-current, continuity, Lorentz force on a charge density

Problem. Define J^μ , write continuity, derive $f_\mu = J^\nu F_{\nu\mu}$, identify f_0 .

4-current:

$$J^\mu = \rho_0 U^\mu = (\rho c, \mathbf{j}), \quad \mathbf{j} = \rho \mathbf{v}.$$

Continuity:

$$\boxed{\partial_\mu J^\mu = 0 \quad \Leftrightarrow \quad \partial_t \rho + \nabla \cdot \mathbf{j} = 0.}$$

Lorentz force per unit volume:

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{j} \times \mathbf{B}.$$

Power per unit volume (rate of work done on charges):

$$f^0 c = \mathbf{j} \cdot \mathbf{E}.$$

Derive $f_\mu = J^\nu F_{\nu\mu}$. Field tensor:

$$F_{\nu\mu} = \partial_\nu A_\mu - \partial_\mu A_\nu, \quad F^{0i} = -E_i/c, \quad F^{ij} = -\epsilon_{ijk} B_k.$$

Compute $J^\nu F_{\nu\mu}$ for $\mu = i$ (spatial):

$$J^\nu F_{\nu i} = J^0 F_{0i} + J^j F_{ji}.$$

Insert components ($F_{0i} = E_i/c$, $F_{ji} = \epsilon_{jik} B_k$, $J^0 = \rho c$, $J^j = j_j$):

$$J^\nu F_{\nu i} = \rho E_i + (\mathbf{j} \times \mathbf{B})_i = f_i.$$

Done.

f^0 **component:**

$$f^0 = J^j F_j^0 = (\mathbf{j} \cdot \mathbf{E})/c.$$

$f^0 = \mathbf{j} \cdot \mathbf{E}/c = \text{rate of work done by the field per unit volume, divided by } c.$

Physically: power density delivered by the EM field to the charges (the time component of the 4-force density).

(c) $F^{\alpha\beta}$ from given A^μ

Problem. $A^\mu = (0, A_0 \cos(K \cdot X), A_0 \sin(K \cdot X), 0)$, $K = (\omega/c, 0, 0, k_z)$. Compute $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$.

Phase: $\Phi \equiv K^\mu X_\mu = \omega t - k_z z$ (using metric $(+, -, -, -)$, $K \cdot X = (\omega/c)(ct) - k_z z$).

Components: $A^1 = A_0 \cos \Phi$, $A^2 = A_0 \sin \Phi$, $A^0 = A^3 = 0$.

Derivatives ($\partial^0 = (1/c)\partial_t$, $\partial^i = -\partial_i$, $\partial_0 \Phi = \omega/c$, $\partial_3 \Phi = -k_z$):

$$\partial^0 A^1 = (1/c)\partial_t A^1 = -A_0(\omega/c) \sin \Phi.$$

$$\partial^0 A^2 = A_0(\omega/c) \cos \Phi.$$

$$\partial^3 A^1 = -\partial_z A^1 = -A_0 k_z \sin \Phi.$$

$$\partial^3 A^2 = A_0 k_z \cos \Phi.$$

$$\partial^1 A^0 = \partial^2 A^0 = \partial^1 A^3 = \partial^2 A^3 = 0.$$

Antisymmetric tensor:

$$F^{01} = \partial^0 A^1 - \partial^1 A^0 = -A_0(\omega/c) \sin \Phi.$$

$$F^{02} = A_0(\omega/c) \cos \Phi.$$

$$F^{31} = \partial^3 A^1 - \partial^1 A^3 = -A_0 k_z \sin \Phi.$$

$$F^{32} = A_0 k_z \cos \Phi.$$

$$F^{12} = \partial^1 A^2 - \partial^2 A^1 = 0.$$

Identify $\mathbf{E}$, $\mathbf{B}$ from $F^{0i} = -E_i/c$, $F^{ij} = -\epsilon_{ijk} B_k$:

$$\mathbf{E} = (A_0 \omega \sin \Phi, -A_0 \omega \cos \Phi, 0).$$

$$\mathbf{B} = (-A_0 k_z \cos \Phi, -A_0 k_z \sin \Phi, 0).$$

$|\mathbf{E}| = A_0 \omega, |\mathbf{B}| = A_0 k_z, |\mathbf{E}| = c|\mathbf{B}|, \mathbf{E} \cdot \mathbf{B} = 0.$

Circularly polarised plane wave propagating along $+\hat{z}$.

(d) Field transformation under boost

Problem. S' moves with 3-velocity $\mathbf{v}$ relative to S . Write $\mathbf{E}'$, $\mathbf{B}'$.

Decompose into components parallel ($\parallel$) and perpendicular ($\perp$) to $\mathbf{v}$:

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad \mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}.$$

$$\boxed{\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B})_{\perp}, \quad \mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - (\mathbf{v} \times \mathbf{E})/c^2)_{\perp}.}$$

(e) Capacitor + magnetic field

Problem. Plates parallel to yz -plane, gap $d = 0.1$ m, $E = 100$ MV/m between plates, electron launched from negative plate at rest. (i) Energy and velocity at the second plate; (ii) can a $\mathbf{B}$ prevent it hitting positive plate; (iii) direction; (iv) magnitude; (v) trajectory sketch.

(i) Energy and velocity at the second plate.

Energy gained:

$$\Delta E = eEd = 100 \text{ MeV/m} \times 0.1 \text{ m} \times e \cdot 1 = 10 \text{ MeV}.$$

Wait: $E = 100$ MV/m $\Rightarrow eE = 100 \text{ MeV/m} \cdot e/e = 100 \text{ MeV/m}$ (in eV-units). Times $d = 0.1$ m:

$$eEd = 10 \text{ MeV}.$$

Total energy:

$$E_{\text{tot}} = m_e c^2 + 10 = 0.511 + 10 = 10.511 \text{ MeV}.$$

$$\gamma = E_{\text{tot}}/m_e c^2 = 10.511/0.511 = 20.57.$$

$$\beta = \sqrt{1 - 1/\gamma^2} = \sqrt{1 - 1/423.1} = \sqrt{0.99764} = 0.99882.$$

$$\boxed{E_{\text{kin}} \approx 10 \text{ MeV}, \quad \gamma \approx 20.6, \quad v \approx 0.999 c.}$$

(ii) Can $\mathbf{B}$ prevent it reaching the positive plate?

Yes. A magnetic field perpendicular to the electron velocity bends its trajectory; if the field is strong enough, the gyroradius is smaller than the gap so that the electron curves back to the negative plate before hitting the positive plate.

$$\boxed{\text{Yes — a transverse } \mathbf{B} \text{ bends the electron back.}}$$

(iii) Direction of $\mathbf{B}$. Take electron initial $\mathbf{v}$ along $+\hat{x}$. To bend it inside the xz (or xy) plane back toward the negative plate, $\mathbf{B}$ must lie in the yz -plane (perpendicular to $\mathbf{v}$). Choosing $\mathbf{B} = B\hat{y}$:

$$\boxed{\mathbf{B} \parallel \hat{y} \text{ (or } \hat{z}\text{): perpendicular to } \mathbf{v} \text{ and to } \mathbf{E}.}$$

The electron then executes (modified) cyclotron motion in the xz -plane.

(iv) Strength of $\mathbf{B}$ to stop the electron at $x = d$.

Energy conservation in static fields ($\mathbf{B}$ does no work):

$$\frac{1}{2}(\text{at any position } x) : \gamma(x)m_e c^2 = m_e c^2 + eEx.$$

At the turn-back point the electron must have $v_x = 0$. Use canonical momentum conservation: with $\mathbf{B} = B\hat{y}$ and $\mathbf{A} = -Bx\hat{z}$ (so $\mathbf{B} = \nabla \times \mathbf{A} = B\hat{y}$), A_z depends on x only, and the system is independent of z :

$$p_z + qA_z = \text{const}, \quad q = -e.$$

At launch: $p_z = 0$, $x = 0 \Rightarrow A_z = 0$:

$$p_z(x) - e(-Bx) = 0 \Rightarrow p_z(x) = -eBx.$$

Hmm sign: with $q = -e$ and $A_z = -Bx$, conservation of $p_z + qA_z$:

$$p_z(x) + (-e)(-Bx) = 0 \Rightarrow p_z(x) = -eBx.$$

At turn-around $v_x = 0$, so $|p_z(x_{\max})| = p_{\text{tot}}(x_{\max}) = \gamma(x_{\max})m_e v(x_{\max})$.

Total momentum at x :

$$p_{\text{tot}}(x)c = \sqrt{E_{\text{tot}}(x)^2 - (m_e c^2)^2}, \quad E_{\text{tot}}(x) = m_e c^2 + eEx.$$

Set $|p_z| = p_{\text{tot}}$ at $x = x_{\max} = d$:

$$eBd = p_{\text{tot}}(d).$$

Solve:

$$B = p_{\text{tot}}(d)/(ed).$$

At $x = d$: $E_{\text{tot}} = 10.511 \text{ MeV}$, so

$$p_{\text{tot}}c = \sqrt{10.511^2 - 0.511^2} = \sqrt{110.48 - 0.261} = \sqrt{110.22} = 10.499 \text{ MeV}.$$

Hence $p_{\text{tot}} = 10.499 \text{ MeV}/c$. Convert:

$$B = \frac{10.499 \text{ MeV}/c}{e \cdot 0.1 \text{ m}} = \frac{10.499 \times 10^6 \text{ eV}}{c \cdot 0.1 \text{ m}} \text{ T-equivalent.}$$

Numerically using $1 \text{ T} \cdot c \cdot 1 \text{ m} = 2.998 \times 10^8 \text{ V}$:

$$B = \frac{pc}{edc} = \frac{10.499 \times 10^6 \text{ V}}{2.998 \times 10^8 \text{ V/T} \cdot \text{m} \times 0.1 \text{ m}}.$$

Compute denominator:

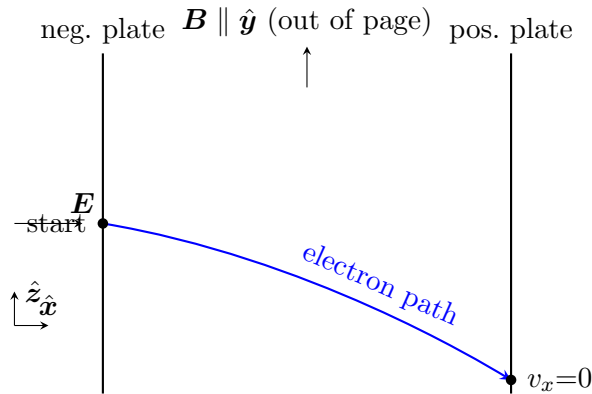
$$2.998 \times 10^8 \times 0.1 = 2.998 \times 10^7 \text{ V/T}.$$

$$B = 10.499 \times 10^6 / 2.998 \times 10^7 = 0.3502 \text{ T}.$$

$$\boxed{B \approx 0.35 \text{ T.}}$$

(v) Trajectory sketch.

The electron starts at the negative plate with $\mathbf{v} = \mathbf{0}$. Initially $\mathbf{B}$ exerts no force; the $\mathbf{E}$ -field accelerates it along $+\hat{x}$. As v_x grows, the magnetic force $-e\mathbf{v} \times \mathbf{B} = -ev_x \hat{x} \times B\hat{y} = -ev_x B\hat{z}$ deflects it toward $-\hat{z}$. The electron curves; if B is exactly the threshold 0.35 T found above, $v_x \rightarrow 0$ at $x = d$ (just touching the positive plate momentarily), then it returns. Trajectory is a cycloidal arc concave toward $-\hat{z}$.



If B slightly exceeds threshold, the electron turns back before $x = d$ and executes a cycloidal drift along $-\hat{z}$ between the plates (drift velocity $\mathbf{v}_d = \mathbf{E} \times \mathbf{B}/B^2$).